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Aim: The goals of the project are as follows:

(1) Implement the Adaptive Controller in Pliego 20 paper (without considering control Allocation matrix).

(2) Modify the controller if the control Allocation matrix is unknown. That is to say, instead of thrust and torque, individual motor forces are what we can control.

System Description:

$$\dot{p} = v$$

$$m\dot{v} = TRz_i - mgz_i$$

$$\dot{R} = RS(\omega)$$

$$J\dot{\omega} = \tau - S(\omega)J(\omega)$$

$$\begin{bmatrix} T \\ \tau \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -a_1 & a_1 & a_1 & -a_1 \\ a_1 & -a_1 & a_1 & -a_1 \\ a_2 & a_2 & -a_2 & -a_2 \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$$

where $\begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$ are the individual motor forces.

The rest of the notations and symbols are exactly the same as those used in the paper, except S_p replaced by z & S_o replaced by z_i for consistency with lecture notes.

Below is an extended and exhaustive explanation of the proposed controller in Pliego 20 & then the modification for control matrix adaptation.

The python notebook has the entire codeblock (with necessary explanations & comments) as well as all the required plots. (It would take some time if you re-run it).

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$\dot{p} = v$, $\dot{p}_d(t) = \dot{v}_d(t)$

$m\dot{v} = TR z_i - mg z_i$

$\tilde{p} = p_d(t) - p$

$\tilde{v} = v_d(t) - v - \dot{\tilde{p}}$

We want both $\tilde{p}$ and $\tilde{v} \rightarrow 0$

$\dot{\tilde{p}} - \tilde{v}$

$m\dot{\tilde{v}} = m\dot{v}_d(t) - m\dot{v} = m\dot{v}_d(t) - \{ TR z_i - mg z_i \}$

$= m(\dot{v}_d(t) + g z_i) - TR z_i$

Assume $\tilde{v}$ to be the control, we would want it to be $-k_p \tilde{p}$.

$z = \tilde{v} - (-k_p \tilde{p}) = \tilde{v} + k_p \tilde{p} \quad \Rightarrow \tilde{v} = z - k_p \tilde{p}$

$m\dot{z} = m(\dot{v}_d(t) + g z_i) - TR z_i + m k_p \tilde{v}$

System is under actuated (only T (scalar) available for control)

In order to get the desired orientation (Q_d or R_d) and the required thrust magnitude, we introduce a virtual control

$$f = TR_d z_i$$

Thus,

$m\dot{z} = m(\dot{v}_d(t) + g z_i + k_p \tilde{v}) - f - T(R - R_d) z_i$ -- (we can take $v_r = v_d + k_p \tilde{p}$)

Regressor $= \Psi_p(\dot{v}_r) = \dot{v}_r + g z_i$

Objective : $z \rightarrow 0$ & $p \rightarrow 0$

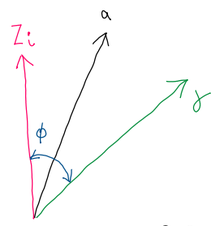
Proposed $f = k_v z + \alpha_p \tilde{p} + \hat{m}(t) \Psi_p(\dot{v}_r)$

& the suggested parameter update $\dot{\hat{m}} = \gamma_p \Psi_p^T(\dot{v}_r) z$

Once we have f, we can find the desired orientation of the body z axis by finding a unit vector along f

$$\gamma = \frac{f}{\|f\|}$$

z_i can be oriented along this by rotating about a vector perpendicular to both by the angle between them.



$$a = \frac{z_i \times \gamma}{\|z_i \times \gamma\|} = \frac{1}{\sqrt{1-\gamma_z^2}} \begin{bmatrix} -\gamma_y \\ \gamma_x \\ 0 \end{bmatrix}$$

& $Q_a = \{\cos(\phi/2), \sin(\phi/2)\, a\}$

Now, $\cos\phi = \frac{z_i \cdot \gamma}{\|z_i\| \|\gamma\|} = \frac{\gamma_z}{(1)(1)} \Rightarrow \cos(\phi/2) = \sqrt{\frac{1+\gamma_z}{2}}$, $\sin(\phi/2) = \sqrt{\frac{1-\gamma_z}{2}}$

$Q_a \odot z_i = \gamma$

To find the corresponding ω, we use

$$\dot{Q}_a = \frac{1}{2} Q_a \otimes \{0, \omega_a\}$$

Since we already know $\dot{Q}_a$ (since we know Q_a) we can invert this expression to get ω_a

$\{0, \omega_a\} = 2 Q_a^{-1} \otimes \dot{Q}_a$

$= 2 \{\cos(\phi/2), -\sin(\phi/2)\, a\} \otimes \dot{Q}_a$

Calculations give $\omega_a = \frac{1}{1+\gamma_z} \begin{bmatrix} -\dot{\gamma}_y(1+\gamma_z) + \gamma_y \dot{\gamma}_z \\ \dot{\gamma}_x(1+\gamma_z) - \gamma_x \dot{\gamma}_z \\ \gamma_y \dot{\gamma}_x - \gamma_x \dot{\gamma}_y \end{bmatrix}$

Thus Q_a and ω_a can be considered as the desired orientation and angular velocity. However, we have so far utilized only 3 of the four available controls, which will make our quadrotor z-axis point in the desired direction so that the thrust takes us to the desired location. We can still perform rotation about the body z-axis (same as performing Yaw in the Euler angle framework)

This would mean $Q_\psi = \{\cos(\psi_d(t)/2),\ \sin(\psi_d(t)/2)\, z_1\}$

which is the quaternion corresponding to the yaw motion

And the overall desired quaternion would then just be

$$Q_d = Q_a \otimes Q_\psi$$

The overall desired angular velocity would satisfy

$$\dot{Q}_d = \frac{1}{2} Q_d \otimes \{0, \omega_d\} \Rightarrow \{0, \omega_d\} = 2 Q_d^{-1} \otimes \dot{Q}_d$$

Now, $\dot{Q}_d = \dot{Q}_a \otimes Q_\psi + Q_a \otimes \dot{Q}_\psi$

$$\therefore \{0, \omega_d\} = 2\{Q_a \otimes Q_\psi\}^{-1} \otimes \{\dot{Q}_a \otimes Q_\psi + Q_a \otimes \dot{Q}_\psi\}$$

$$= 2\{Q_\psi^{-1} \otimes \underbrace{Q_a^{-1} \otimes \dot{Q}_a}_{\frac{1}{2}\{0,\omega_a\}} \times Q_\psi\} + 2\{Q_\psi^{-1} \otimes \underbrace{Q_a^{-1} \otimes Q_a}_{I} \otimes \dot{Q}_\psi\}$$

$$- Q_\psi^{-1} \otimes \{0, \omega_a\} \otimes Q_\psi + \{0, \omega_\psi\}$$

Thus, $\omega_d = Q_\psi^{-1} \odot \omega_a + \omega_\psi$

Now that we have the desired orientation, we can compute the error quaternion and error in angular velocity (from Homework 4)

$$\tilde{Q} = Q^{-1} \otimes Q_d = \{\tilde{\eta}, \tilde{\epsilon}\},\ \tilde{\omega} = \tilde{Q} \odot \omega_d(t) - \omega$$
$$= R_d \omega_d - \omega$$

$$\dot{\tilde{Q}} = L(\tilde{\eta}, \tilde{\epsilon})\,\tilde{\omega} = \begin{bmatrix} \dot{\tilde{\eta}} \\ \dot{\tilde{\epsilon}} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -\hat{\epsilon}^T \\ \eta I + S(\epsilon) \end{bmatrix} \tilde{\omega}$$

$$J\dot{\tilde{\omega}} = J\left(\dot{R}_d \omega_d + R_d \dot{\omega}_d\right) - \left(\tau - \omega \times J\omega\right)$$

Similar to backstepping part of Homework 4, we assume $\tilde{\omega}$ to be a virtual control first and we want it to be $-K_\epsilon \tilde{\epsilon}$ to drive the error quaternion to 0, and then introduce

$$z_1 = \tilde{\omega} - (-K_\epsilon \tilde{\epsilon}) = \tilde{\omega} + K_\epsilon \tilde{\epsilon}$$

Additionally, now since the two systems (translational and rotational) are coupled, and we want one single Lyapunov function proving convergence of all errors to zero, the paper adds another term into z_1 based on z. We will later see in the analysis how this helps. Thus, overall z_1 is taken to be,

$$z_1 = \tilde{\omega} + K_\epsilon \tilde{\epsilon} + 2\kappa_0^{-1} J S(\bar{\epsilon}) Q^{-1} \odot z$$

We take (just for simplicity) $\omega_r = \tilde{Q} \odot \omega_d(t) + K_\epsilon \tilde{\epsilon} + 2\kappa_0^{-1} J S(\bar{\epsilon}) Q^{-1} \odot z$

Thus, $z_1 = \omega_r - \omega$

Now, $J\dot{z}_1 = J\dot{\omega}_r - J\dot{\omega} = J\omega_r - \{\tau - \omega \times J\omega\}$

$$= J\dot{\omega}_r + \omega_r \times J\omega - \omega_r \times J\omega + \omega \times J\omega - \tau$$

$$= \underbrace{\left(J\dot{\omega}_r + \omega_r \times J\omega\right)}_{\text{Regressor Matrix} = Y_0(\omega, \omega_r, \dot{\omega}_r)\theta(t)} - z_1 \times J\omega - \tau$$

Objective: $\tilde{Q} \to 1_Q$, $\tilde{\omega} \to 0$

The following control law is proposed

$$\tau = K_\omega s_0 + \kappa_0 \tilde{\epsilon} + Y(\omega, \omega_r, \dot{\omega}_r)\hat{\theta}(t)$$

And the following update law is proposed for the parameters $\hat{\theta}$

$$\dot{\hat{\theta}} = \gamma_0 Y_0^T(\omega, \omega_r, \dot{\omega}_r) z_1$$

Here $\theta = \begin{bmatrix} J_{11} \\ J_{22} \\ J_{33} \end{bmatrix}$

and $Y_0(\omega, \omega_r, \dot{\omega}_r)\theta(t)$

$$= J\dot{\omega}_r + \omega_r \times J\omega$$

$$= \begin{bmatrix} J_{11}\dot{\omega}_r[0] - \omega_r[2]\, J_{22}\, \omega[1] + \omega_r[1]\, J_{33}\, \omega[2] \\ J_{22}\dot{\omega}_r[1] - \omega_r[0] \cdot J_{33} \cdot \omega[2] + \omega_r[2] \cdot J_{11}\, \omega[0] \\ J_{33}\omega_r[2] + \omega_r[0]\, J_{22} \cdot \omega[1] - \omega_r[1] \cdot J_{11} \cdot \omega[0] \end{bmatrix}$$

From here, we can get

$$Y(\omega, \omega_r, \dot{\omega}_r) = \begin{bmatrix} \omega_r[0] & -\omega_r[2]\,\omega[1] & \omega_r[1]\,\omega[2] \\ \omega_r[2] \cdot \omega[0] & \omega_r[1] & -\omega_r[0] \cdot \omega[2] \\ -\omega_r[1]\,\omega[0] & \omega_r[0] \cdot \omega[1] & \dot{\omega}_r[2] \end{bmatrix}$$

Stability Analysis

Re-iterating all the relevant relations

$$\tilde{v} = -k_p \tilde{\rho} + z = \dot{\tilde{\rho}}$$
$$\dot{mz} = -k_v z - \alpha_p \tilde{\rho} - \tilde{m}\Psi_p(v_s) - T(R - R_d) z_1$$
$$\dot{\hat{m}} = \gamma_p \Psi_p^T(v_p) z$$

$$\tilde{m} = \hat{m}(t) - m$$

$$\dot{\tilde{Q}} = L(\tilde{\eta}, \tilde{\epsilon})\tilde{\omega}$$
$$\tilde{\omega} = -k_\epsilon \tilde{\epsilon} - 2\alpha_0^{-1} T S(\tilde{\epsilon}) Q^{-1} \odot z + z_1$$
$$J z_1 = -z_1 \times J\omega - k_\omega z_1 - \alpha_0 \tilde{\epsilon} - \Psi_0(\omega, \omega_s, \dot{\omega}_s)\tilde{\theta}$$
$$\dot{\hat{\theta}} = \gamma_0 \Psi_0^T(\omega, \omega_s, \dot{\omega}_s) z_1$$

Define the overall Lyapunov function as follows

$$V = \frac{1}{2}\alpha_p \|\tilde{\rho}\|^2 + \frac{1}{2} m \|z\|^2 + \frac{1}{2\gamma_p}\tilde{m}^2 + \alpha_0 \underbrace{\{\|\tilde{\epsilon}\|^2 + (1-\tilde{\eta})^2\}}_{\to 2(1-\tilde{\eta})} + \frac{1}{2} z_1^T J z_1 + \frac{1}{2\gamma_0}\|\tilde{\theta}\|^2$$

$$\dot{V} = \alpha_p \tilde{\rho}^T \dot{\tilde{\rho}} + m z^T \dot{z} + \frac{1}{\gamma_p}\tilde{m}\dot{\tilde{m}} + \alpha_0 2(-\dot{\tilde{\eta}}) + z_1^T J \dot{z}_1 + \frac{1}{\gamma_0}\tilde{\theta}\dot{\tilde{\theta}}$$

$$= \alpha_p \tilde{\rho}^T(-k_p\tilde{\rho} + z) + z^T(-k_v z - \alpha_p \tilde{\rho} - \tilde{m}\Psi_p(v_s) - T(R - R_d) z_1)$$
$$+ \frac{1}{\gamma_p}\tilde{m}\left(\gamma_p \Psi_p^T(v_s) z\right) - 2\alpha_0\left(-\frac{1}{2}\tilde{\epsilon}^T\tilde{\omega}\right)$$
$$+ z_1^T\left(-z_1 \times J\omega - k_\omega z_1 - \alpha_0\tilde{\epsilon} - \Psi_0(\omega, \omega_s, \dot{\omega}_s)\tilde{\theta}\right)$$
$$+ \frac{1}{\gamma_0}\tilde{\theta}\gamma_0 \Psi_0^T(\omega, \omega_s, \dot{\omega}_s) z_1$$

$$= -\alpha_p k_p \|\tilde{\rho}\|^2 - k_v \|z\|^2 - z^T T(R - R_d) z_1$$
$$+ \alpha_0 \tilde{\epsilon}^T\tilde{\omega} - z_1^T S(z_1) J\omega - k_\omega \|z_1\|^2 - \alpha_0 z_1^T \tilde{\epsilon}$$

Terms to be managed. $-z_1^T T(R - R_d) z_i + \alpha_0 \tilde{\epsilon}^T\tilde{\omega} - \underline{z_1^T S(z_1) J\omega} - \alpha_0 z_1^T\tilde{\epsilon}$

$\to 0$ since $\underline{z_1^T \times z_1 = 0}$.

Now, $(R - R_d) z_i = R(z_i - \tilde{R} z_i) = Q \odot (z_i - \tilde{Q} \odot z_i)$

(since $\tilde{R} = R^T R_d$)

Now, $z_i - \tilde{Q} \odot z_i = \begin{bmatrix}0\\0\\1\end{bmatrix} - \tilde{Q} \otimes \begin{bmatrix}0\\0\\1\end{bmatrix} \otimes \tilde{Q}^{-1}$

$$= \begin{bmatrix}0\\0\\1\end{bmatrix} - 2\begin{bmatrix}\tilde{\eta}\tilde{\epsilon}_y + \tilde{\epsilon}_x\tilde{\epsilon}_z\\ -\tilde{\eta}\tilde{\epsilon}_x + \tilde{\epsilon}_y\tilde{\epsilon}_z\\ \tilde{\epsilon}_x^2 + \tilde{\epsilon}_y^2\end{bmatrix}$$

$$= 2\begin{bmatrix}-\tilde{\epsilon}_x\tilde{\epsilon}_z - \tilde{\eta}\tilde{\epsilon}_y\\ -\tilde{\epsilon}_y\tilde{\epsilon}_z + \tilde{\eta}\tilde{\epsilon}_x\\ \frac{1}{2} - \tilde{\epsilon}_x^2 - \tilde{\epsilon}_y^2\end{bmatrix} = 2S(\bar{\epsilon})\tilde{\epsilon}$$

$$\tilde{Q} \odot z_i = R(\tilde{\eta}, \tilde{\epsilon}) z_i$$

$$R(\tilde{\eta}, \tilde{\epsilon}) = (\tilde{\eta}^2 - \tilde{\epsilon}^T\tilde{\epsilon}) I + 2\tilde{\epsilon}\tilde{\epsilon}^T + 2\tilde{\eta} S(\tilde{\epsilon})$$

Multiplying by z_i gives the third column

which is $= \begin{bmatrix}2\tilde{\eta}\tilde{\epsilon}_y + 2\tilde{\epsilon}_x\tilde{\epsilon}_z\\ -2\tilde{\eta}\tilde{\epsilon}_x + 2\tilde{\epsilon}_y\tilde{\epsilon}_z\\ 2\tilde{\epsilon}_z^2 + \tilde{\eta}^2 - \tilde{\epsilon}_x^2 - \tilde{\epsilon}_y^2 - \tilde{\epsilon}_z^2\end{bmatrix}$

where we define $\bar{\epsilon} = \begin{bmatrix}\tilde{\epsilon}_y\\ -\tilde{\epsilon}_x\\ \tilde{\eta}\end{bmatrix}$

Thus, we have $-2z^T T\left(Q \odot S(\bar{\epsilon})\tilde{\epsilon}\right) + \alpha_0\left(\tilde{\epsilon}^T\right)\left(-k_\epsilon\tilde{\epsilon} - 2\alpha_0^{-1} T S(\bar{\epsilon}) Q^{-1} \odot z + z_1\right) - \alpha_0 z_1^T\tilde{\epsilon}$

$$= -k_\epsilon \alpha_0 \|\tilde{\epsilon}\|^T + 2T\left(-z^T Q \odot \delta(\bar{\epsilon})\tilde{\epsilon} - \tilde{\epsilon}^T S(\bar{\epsilon}) Q^{-1} \odot z\right)$$

$$= 0$$

Hence, $\dot{V} = -\alpha_p k_p \|\tilde{\rho}\|^2 - \alpha_0 k_\epsilon \|\epsilon\|^2 - k_v \|z\|^2 - k_\omega \|z_1\|^2 \le 0$

Control Allocation Unknown.

As mentioned in Wang (6), the control allocation matrix has the following form

$$\begin{bmatrix}T\\ \tau_x\\ \tau_y\\ \tau_z\end{bmatrix} = \begin{bmatrix}1 & 1 & 1 & 1\\ -l/\sqrt{2} & l/\sqrt{2} & l/\sqrt{2} & -l/\sqrt{2}\\ l/\sqrt{2} & -l/\sqrt{2} & l/\sqrt{2} & -l/\sqrt{2}\\ c_M/c_T & c_M/c_T & -c_M/c_T & -c_M/c_T\end{bmatrix}\begin{bmatrix}F_1\\ F_2\\ F_3\\ F_4\end{bmatrix}$$

we can take the ratio $c_M/c_T = a_2 l/\sqrt{2} = a_1$, because that is what appears in the matrix and write

$$\begin{bmatrix}T\\ \tau_x\\ \tau_y\\ \tau_z\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & a_1 & 0 & 0\\ 0 & 0 & a_1 & 0\\ 0 & 0 & 0 & a_2\end{bmatrix}\begin{bmatrix}1 & 1 & 1 & 1\\ -1 & 1 & 1 & -1\\ 1 & -1 & 1 & -1\\ 1 & 1 & -1 & -1\end{bmatrix}\begin{bmatrix}F_1\\ F_2\\ F_3\\ F_4\end{bmatrix}$$

As can be seen, there are no unknowns in the translational control T and we can write

$$T = F_1 + F_2 + F_3 + F_4$$

And, if a_1 and a_2 are not known, then

$$\begin{bmatrix} T \\ \tau \end{bmatrix} = \begin{bmatrix} T \\ \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & a_1 & 0 & 0 \\ 0 & 0 & a_1 & 0 \\ 0 & 0 & 0 & a_2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix}}_{B} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$$

$$\left\{ \text{diag}\left(1, a_1, a_1, a_2\right) \right\}^{-1} = \underbrace{\text{diag}\left(1, \frac{1}{a_1}, \frac{1}{a_1}, \frac{1}{a_2}\right)}_{\Lambda_0} \quad \text{---} \quad \text{assuming } a_1, a_2 \neq 0$$

$$\det(B) = \begin{vmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 2 & -2 \\ 2 & 0 & 0 & -2 \\ 0 & 0 & -1 & -1 \end{vmatrix} = (-1) \begin{vmatrix} 0 & 2 & -2 \\ 2 & 0 & -2 \\ 0 & -1 & -1 \end{vmatrix} = (2)(-2-2) \neq 0$$

Thus, B is invertible

The actual control is $F = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix} = B^{-1} \Lambda_0 \begin{bmatrix} T \\ \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix}$

Since Λ_0 is unknown, we cannot get the F corresponding to our desired torque values.

Thus, we will not be able to apply the desired torque that we found in the preceding section.

We thus use an estimate of Λ_0 (which we will call $\hat{\Lambda}_0$) and we now calculate F that will be required based on our estimate of Λ_0 ($\hat{\Lambda}_0$) and the desired torques.

$$F = B^{-1} \hat{\Lambda}_0 \left(\begin{bmatrix} T_d \\ \tau_d \end{bmatrix} \right) \quad \text{----} \quad \left\{ \text{Thus is what we will control. We cannot directly control the torque} \right\}$$

Thus, the torque that will be produced is

$$\begin{bmatrix} T \\ \tau \end{bmatrix} = \Lambda_0^{-1} B \left(B^{-1} \hat{\Lambda}_0 \begin{bmatrix} T_d \\ \tau_d \end{bmatrix} \right)$$

$$= \Lambda_0^{-1} \hat{\Lambda}_0 \begin{bmatrix} T_d \\ \tau_d \end{bmatrix}$$

Λ_0 and $\hat{\Lambda}_0$ are both diagonal and the first element is 1. Thus, thrust value is not affected. We have

$$\tau = \begin{bmatrix} a_1 & 0 & 0 \\ 0 & a_1 & 0 \\ 0 & 0 & a_2 \end{bmatrix} \begin{bmatrix} 1/\hat{a}_1 & 0 & 0 \\ 0 & 1/\hat{a}_1 & 0 \\ 0 & 0 & 1/\hat{a}_2 \end{bmatrix} \tau_d \quad \text{---} \left\{ \text{We denote } \Lambda_0 = \text{diag}\left(\frac{1}{a_1}, \frac{1}{a_1}, \frac{1}{a_2}\right) \text{ here onwards} \right\}$$

We will take our estimates to be $\hat{\rho}_1 = 1/\hat{a}_1$, $\hat{\rho}_2 = 1/\hat{a}_1$ & $\hat{\rho}_3 = 1/\hat{a}_2$

to keep things simple and denote by $\hat{\rho} = [\hat{\rho}_1, \hat{\rho}_2, \hat{\rho}_3]$

Thus, $\tau = \Lambda_0^{-1} \text{diag}(\hat{\rho}) \tau_d$ ---- $\left\{ \text{Thus is not what we control. We control only the motor forces. This is what gets produced from the motor forces that we applied} \right\}$.

In all the equations above, we have used τ_d.

But now we will have to use τ.

We can write it as $\tau = (\tau - \tau_d) + \tau_d$

& the desired thrust can already be matched

Thus, all things remain the same except that we have an additional $(\tau - \tau_d)$ now.

$$\tau - \tau_d = \Lambda_0^{-1}\,\mathrm{diag}\left(\hat{\rho}\right)\tau_d - \Lambda_0^{-1}\,\mathrm{diag}\left(\rho\right)\tau_d$$

$$= \Lambda_0^{-1}\,\mathrm{diag}\left(\tilde{\rho}\right)\tau_d$$

where $\mathrm{diag}\left(\rho\right) = \Lambda_0$ & $\tilde{\rho} = \hat{\rho} - \rho$.

- τ appears in $J\dot{z}_1$, thus $J\dot{z}_1$ remains the same as above except an additional $-\left(\tau - \tau_d\right) = -\Lambda_0^{-1}\,\mathrm{diag}\left(\tilde{\rho}\right)\tau_d$

Thus, in the Lyapunov term corresponding to $z_1 \to \left(\frac{1}{2} z_1^T J z_1\right)$, we will be left with $z_1^T\left(-\Lambda_0^{-1}\,\mathrm{diag}\left(\tilde{\rho}\right)\tau_d\right)$ and the rest of the terms remain as previously

We take the new Lyapunov function,

$V_T = V_{\text{earlier}} + \frac{1}{2}\tilde{\rho}^T \Gamma_\rho^{-1} \Lambda_0^{-1} \tilde{\rho}$ for some positive definite Γ_ρ {we can take identity here}

$\dot{V}_T = \dot{V}_{\text{earlier}} - z_1^T\left(\Lambda_0^{-1}\,\mathrm{diag}\left(\tilde{\rho}\right)\tau_d\right) + \tilde{\rho}^T \Lambda_0^{-1} \dot{\hat{\rho}}$

we can write $\mathrm{diag}\left(\tilde{\rho}\right)\tau_d = \mathrm{diag}\left(\tau_d\right)\tilde{\rho}$

Thus, extra terms $= -z_1^T \Lambda_0^{-1}\,\mathrm{diag}\left(\tau_d\right)\tilde{\rho} + \tilde{\rho}^T \Lambda_0^{-1} \dot{\hat{\rho}}$

Since $\Lambda_0^{-1}, \mathrm{diag}(\tau_d)$ are both diagonal matrices, they commute & $\Lambda_0^{-1}\,\mathrm{diag}\left(\tau_d\right) = \mathrm{diag}\left(\tau_d\right)\Lambda_0^{-1}$

Thus, extra terms $= -z_1^T\,\mathrm{diag}\left(\tau_d\right)\Lambda_0^{-1}\tilde{\rho} + \tilde{\rho}^T \Lambda_0^{-1} \dot{\hat{\rho}}$

$= -\tilde{\rho}^T \Lambda_0^{-1}\,\mathrm{diag}\left(\tau_d\right) z_1 + \tilde{\rho}^T \Lambda_0^{-1} \dot{\hat{\rho}}$ -- $\left(\left(\Lambda_0^{-1}\right)^T = \Lambda_0^{-1}\right)$

Take $\dot{\hat{\rho}} = \mathrm{diag}\left(\tau_d\right) z_1$